



Shiretechnik Solutions Pvt Ltd

SHape Your Idea To Reality

Case Study

on

Thermal Management of Electrical Enclosure



INTRODUCTION

Thermal Management of Electrical Enclosures



Thermal management of electrical enclosures is a design process of controlling internal temperatures & enclosure surface temperature to prevent sensitive components from overheating, failing, or suffering from condensation. It involves evaluating the thermal design and deploying appropriate cooling methods to prolong equipment lifespan.

Effective cooling methods include

- Optimizing Internal Layout (Passive Cooling)
- Sizing & Placing the Air vents
- Closed-loop Cooling for Sealed or IP rated systems
- External Heatsinks on enclosure
- Filtered Fans & Vent Arrangements
- Advanced Cooling Solutions for Higher Power Density



CASE STUDY

Thermal Analysis of Electrical Enclosure

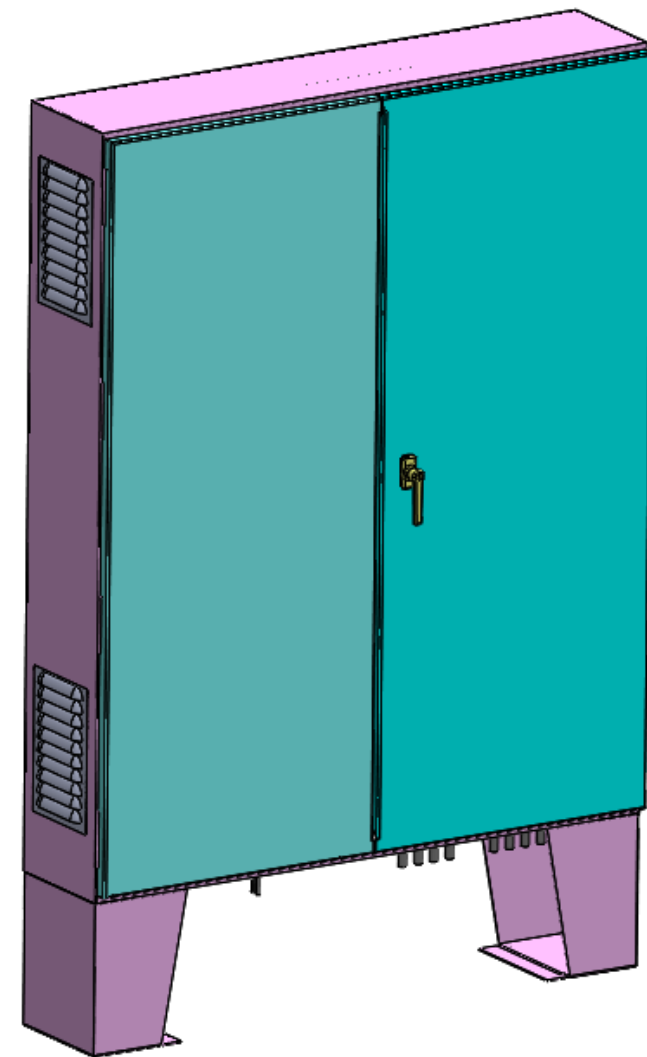


Objective of Thermal Analysis

To evaluate the thermal performance of an industrial electrical enclosure whether the internal components operate within safe temperature limits and further improvise the design for a maximum thermal safety margin.

Thermal Design Overview

Product Type	: Industrial Electrical Enclosure
Key Components	: Power Filters, BR Breakers, Relays, PCB Assemblies, Bus Bars & Cables.
Heat Dissipation	: 120.65 Watt
Ambient Temperature	: 45°C
Solar Load	: Included
Cooling Mode	: Natural Cooling
Enclosure Overall Size	: 1827 x 1220 x 284 mm
Enclosure Material	: Mild Steel
Mounting/Use Type	: Floor Mounted

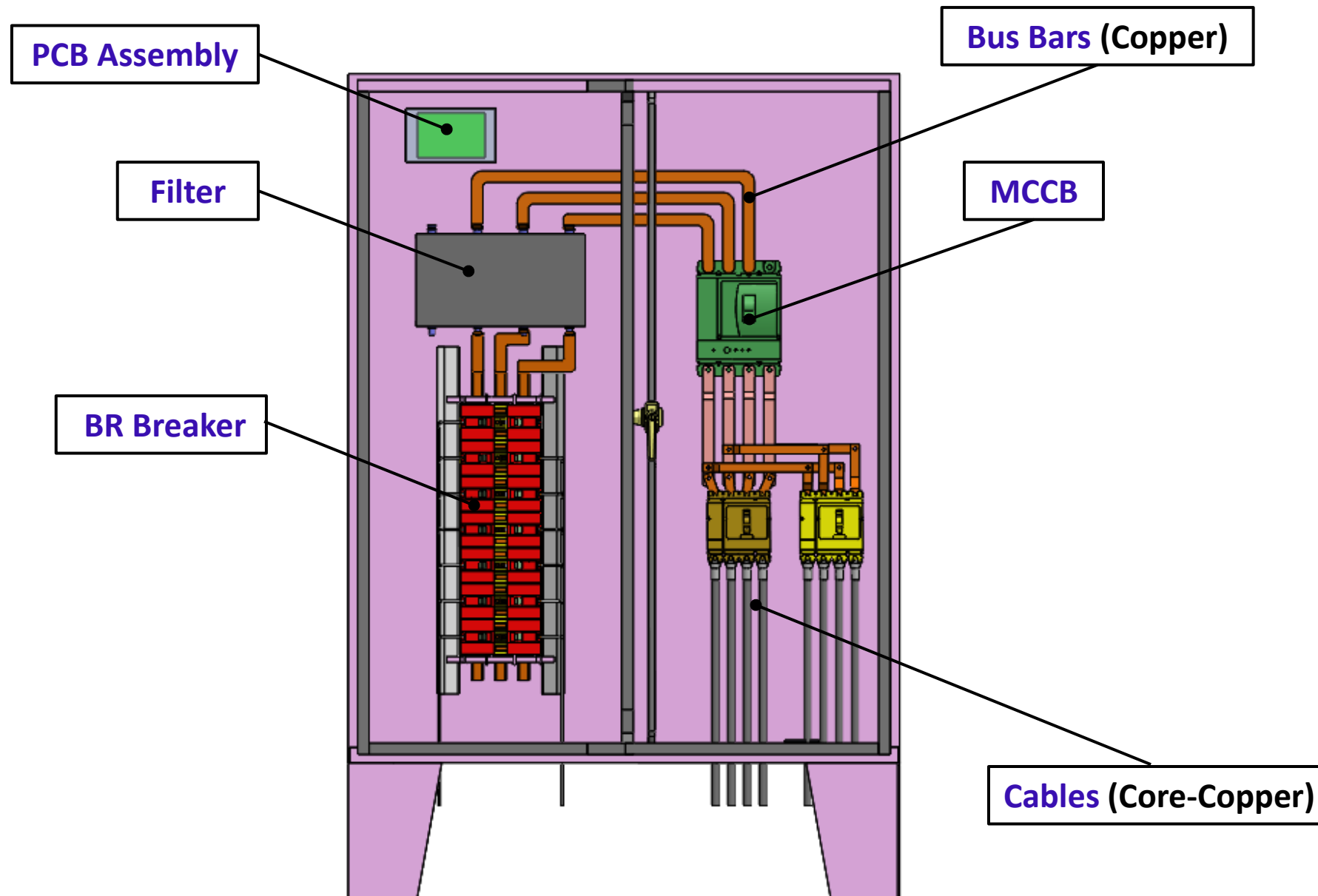


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Enclosure Internal Configuration



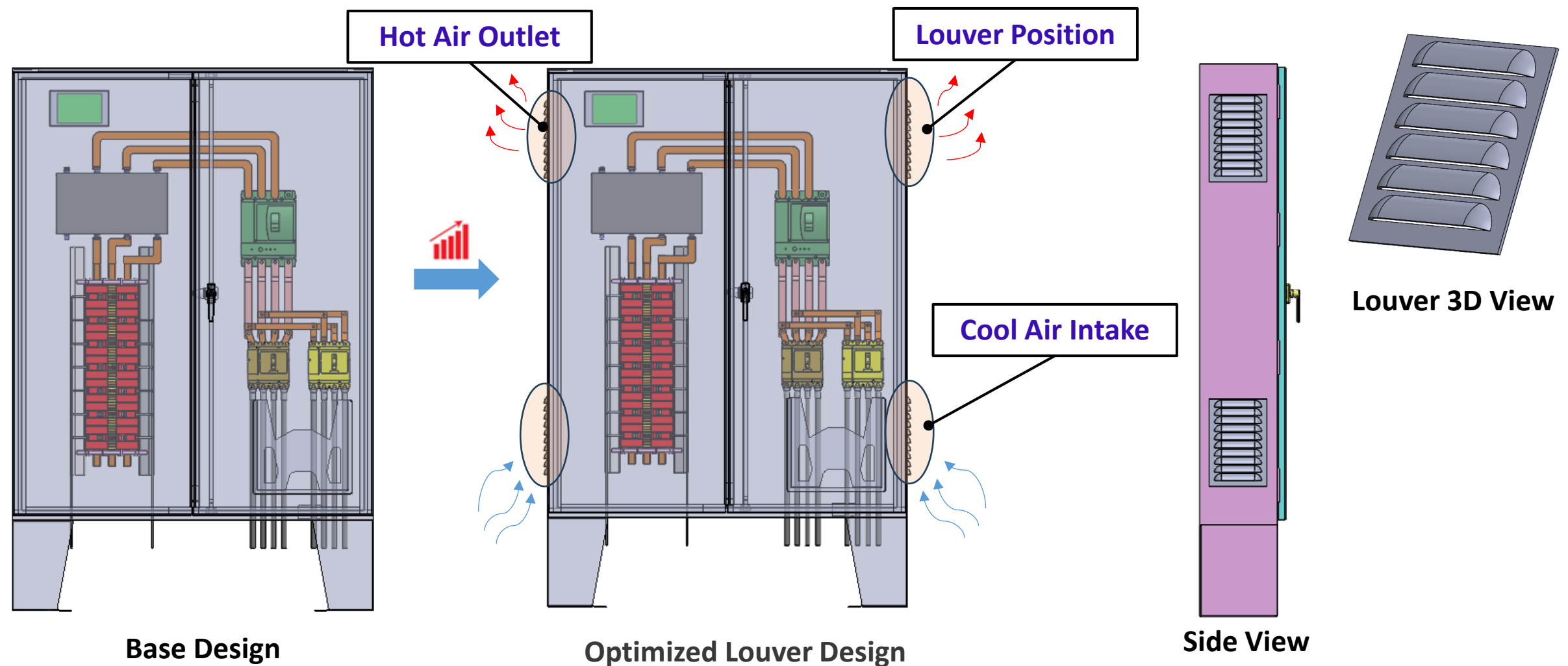
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Optimized Air Vent / Louver Design

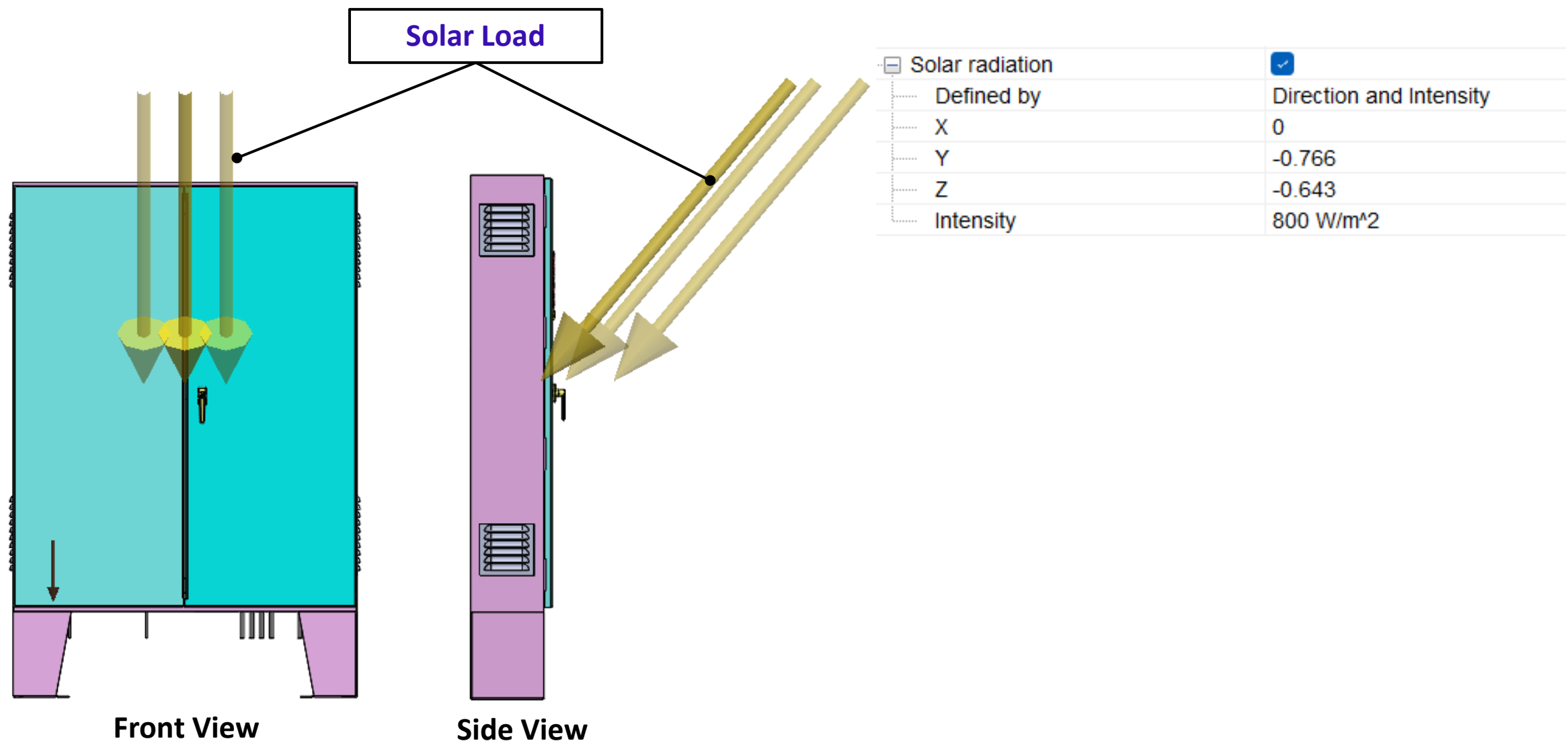
Strategically installed the louver placement on enclosure configuration with which achieved a significant natural sweep pattern & airflow rate, and promoted a maximum heat dissipation from internal power and control components. The results compared against the base design.



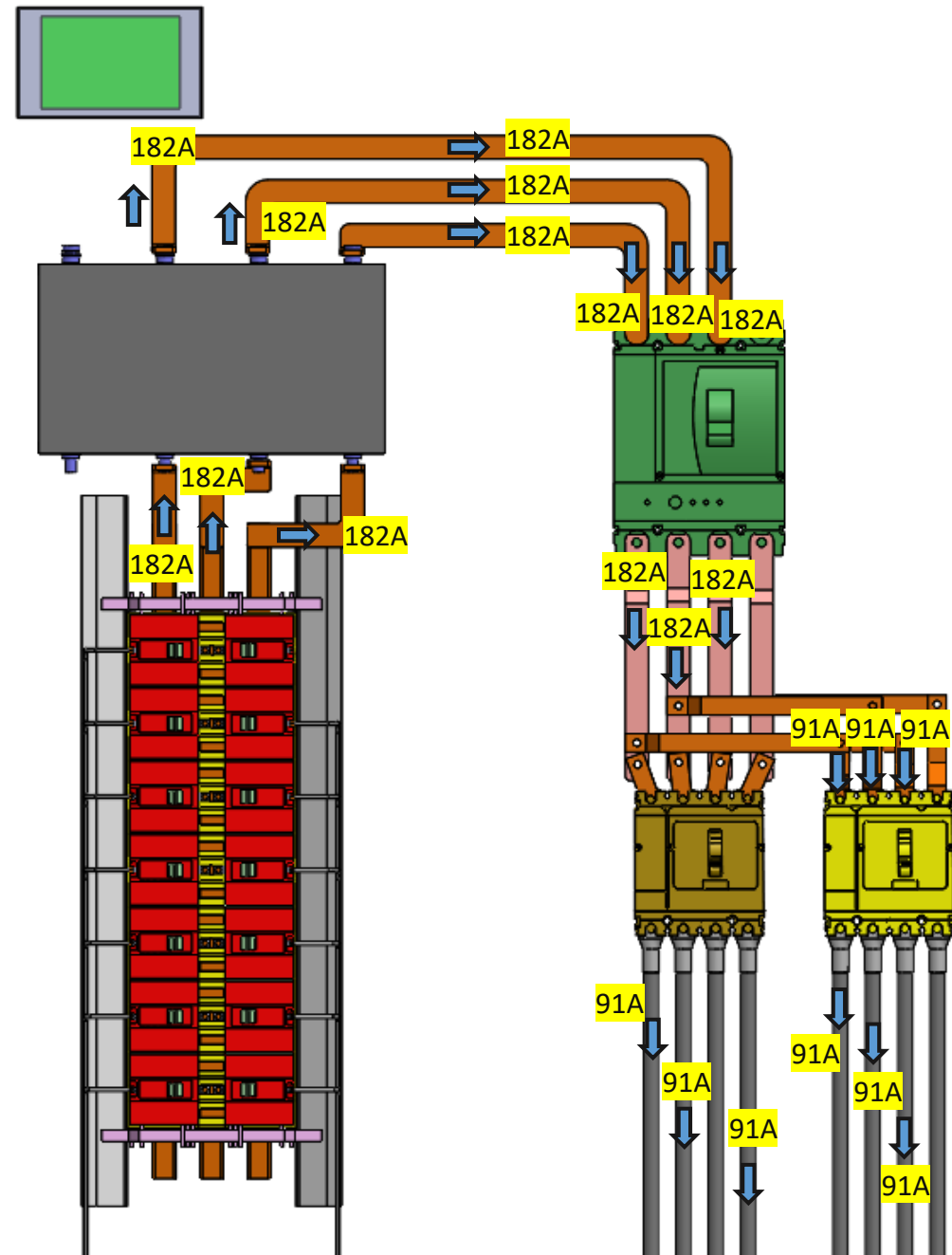
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Solar Radiation

Solar load is included in both configurations to simulate realistic outdoor operating conditions and accurately assess the enclosure thermal performance.



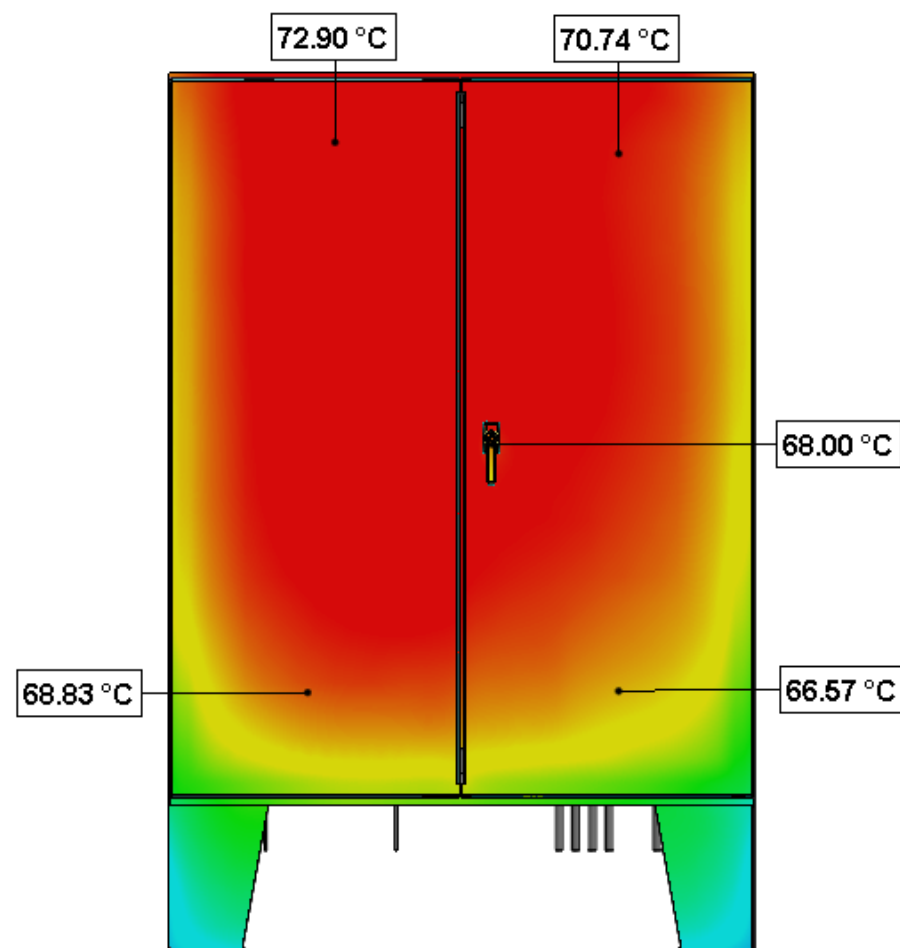
Electric Current Schematic



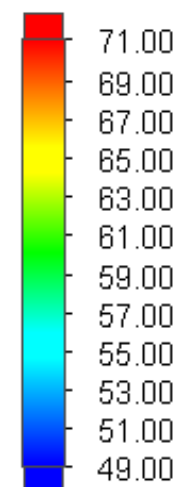
CASE STUDY | Simulation Results

Surface Temperature Contours

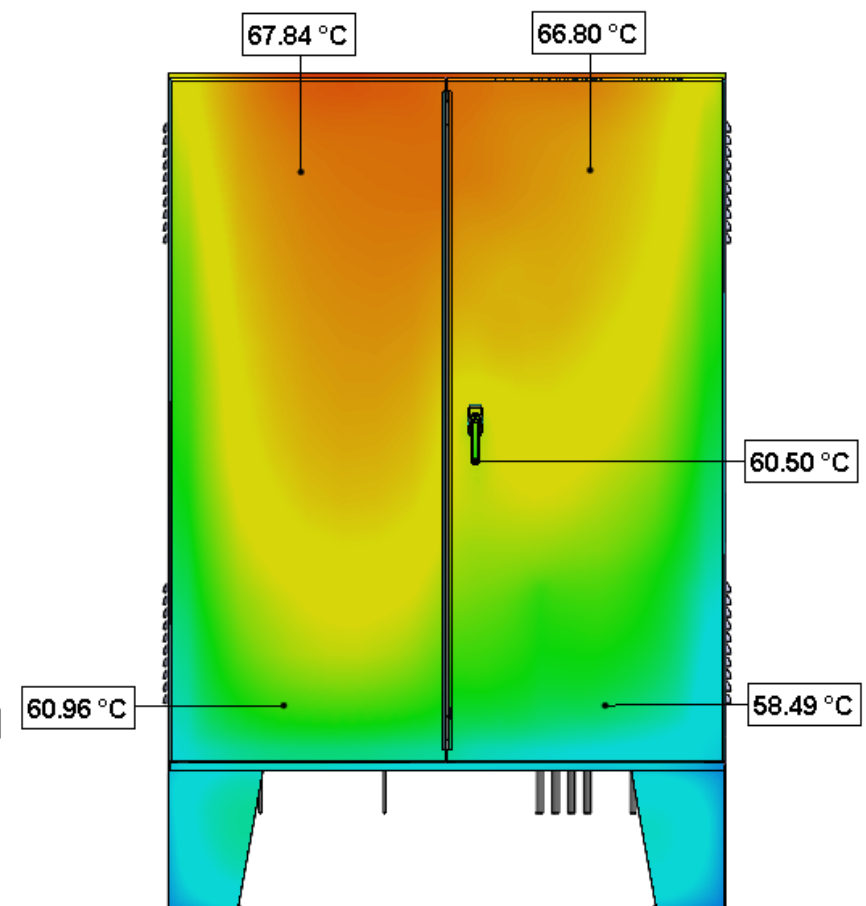
Significant thermal improvement on enclosure surface achieved with the optimized design.



Base Design



Temperature (Solid) [°C]

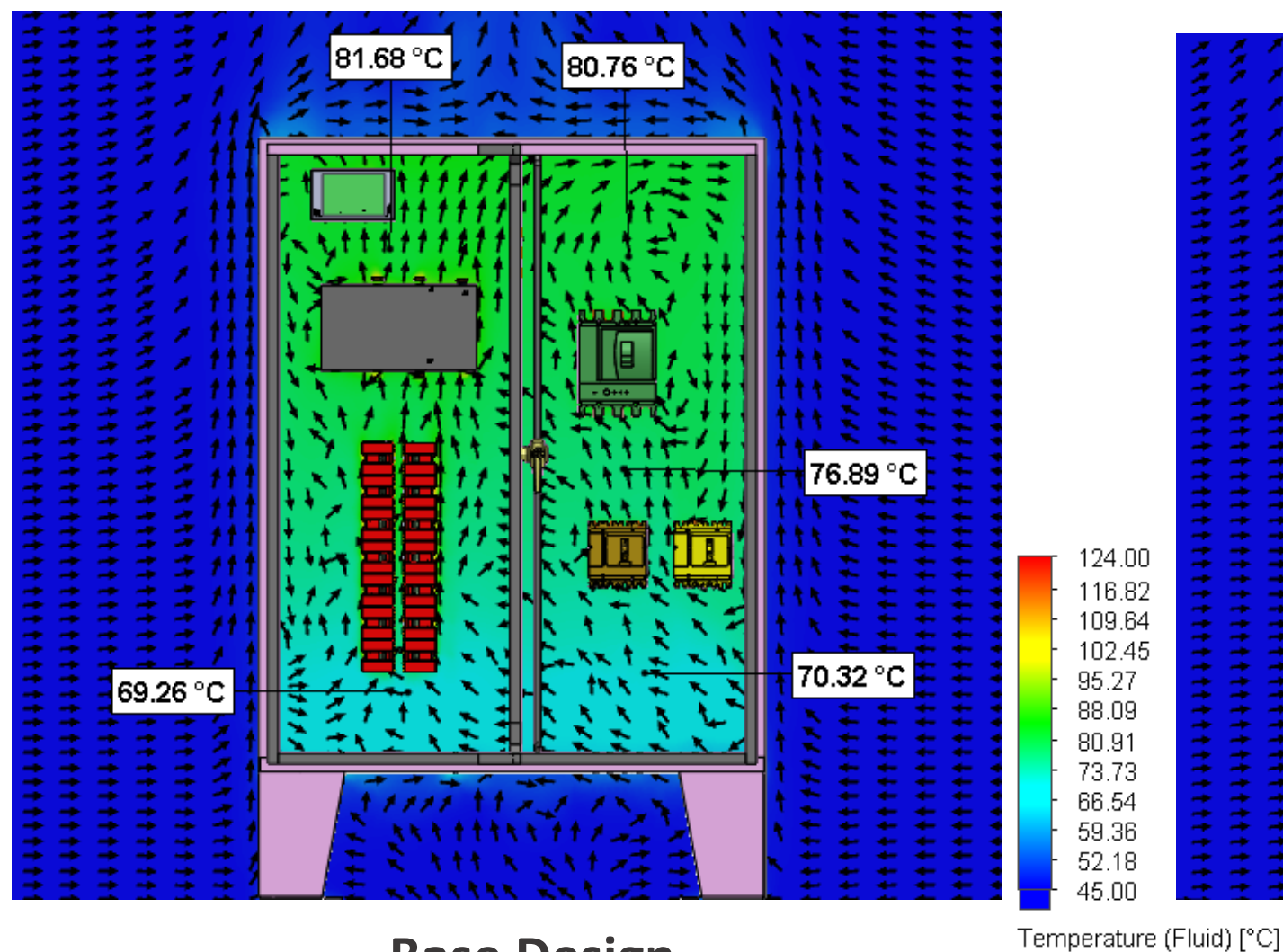


Optimized Design

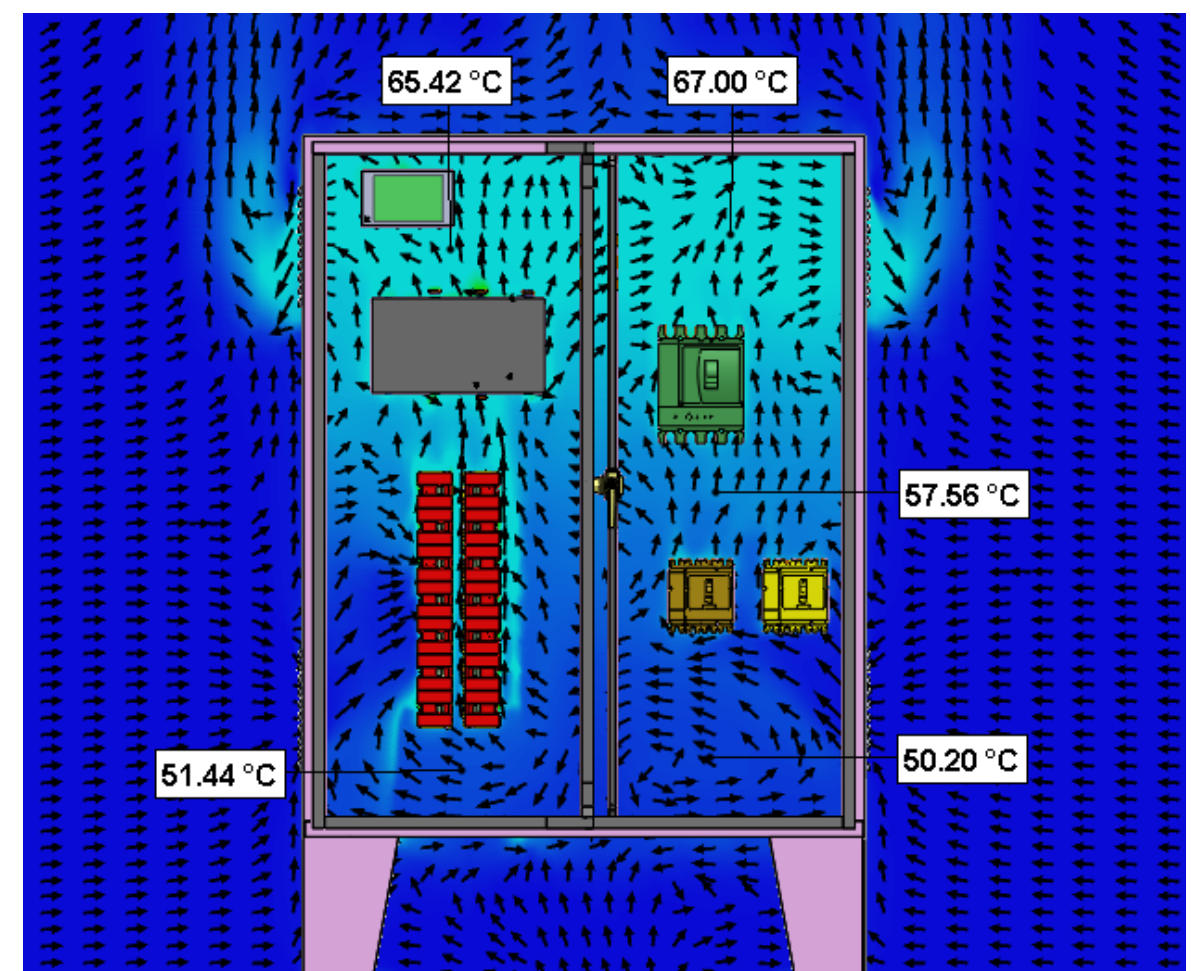
CASE STUDY | Simulation Results

Fluid Temperature Contour with Flow Vectors

Airflow in optimized design shows a well-vented pattern, hence a major reduction on fluid temperature.



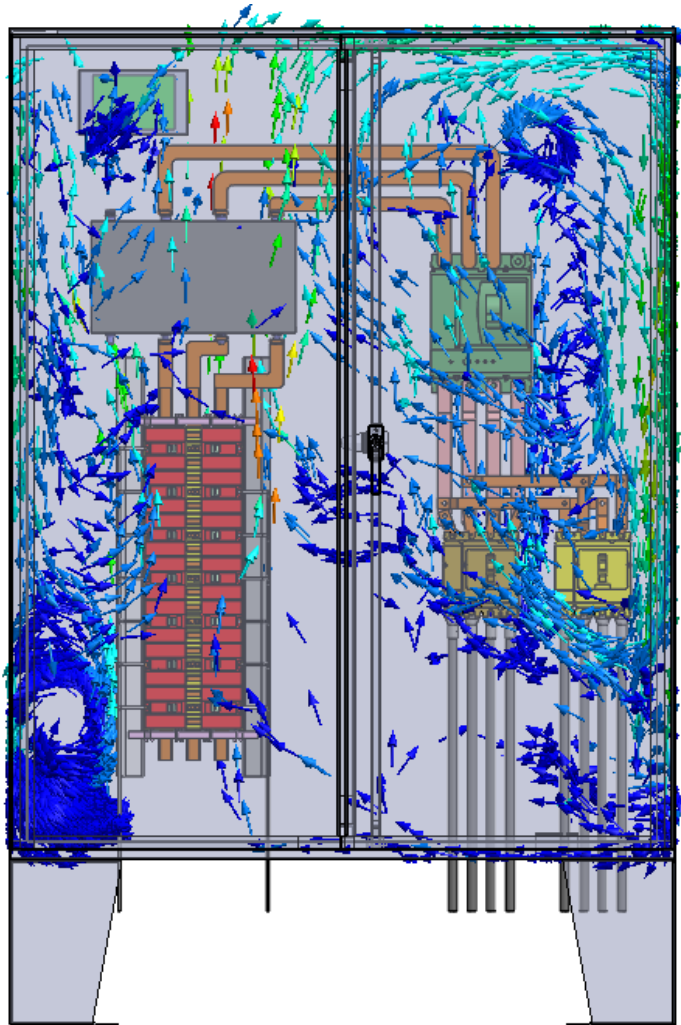
Base Design



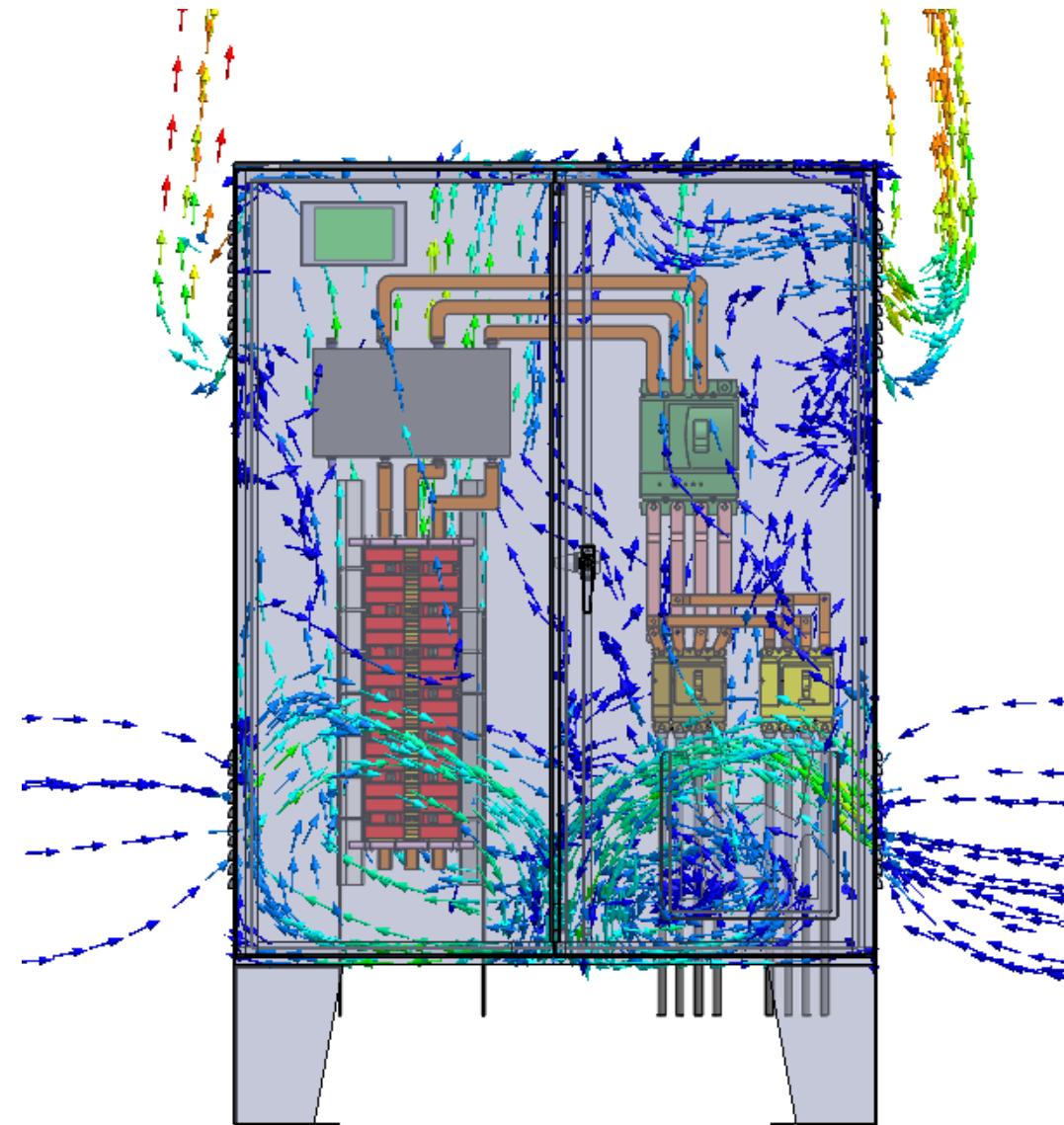
Optimized Design

CASE STUDY | Simulation Results

Airflow Trajectories



Base Design

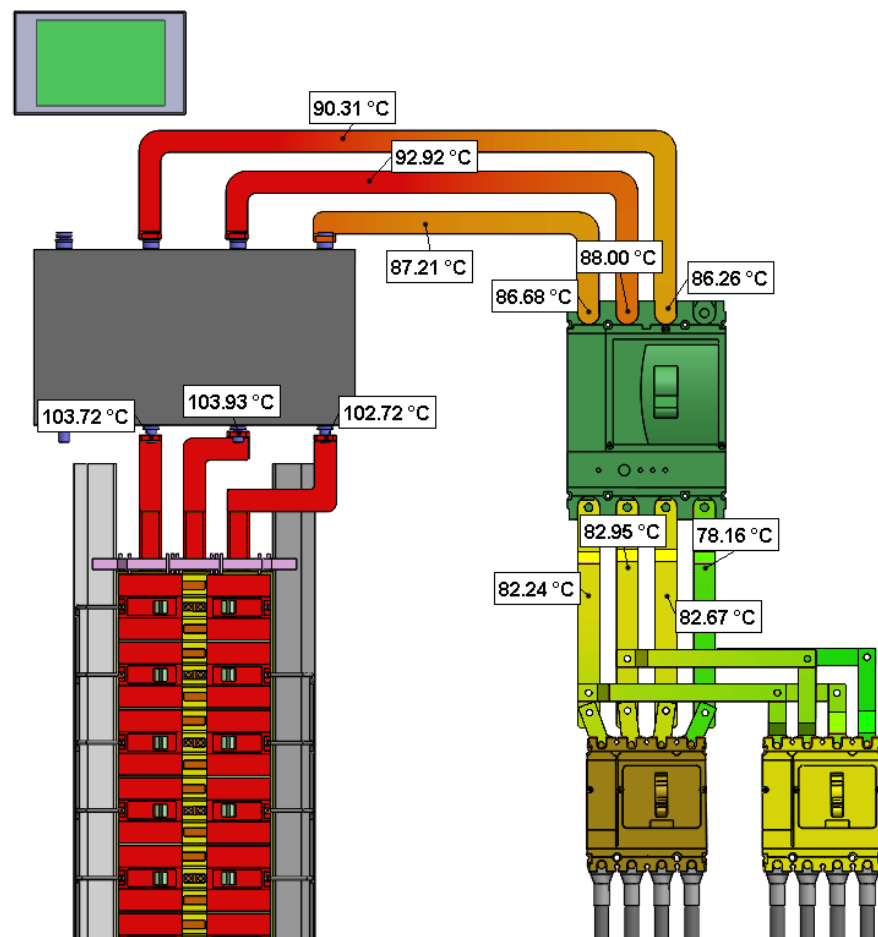


Optimized Design

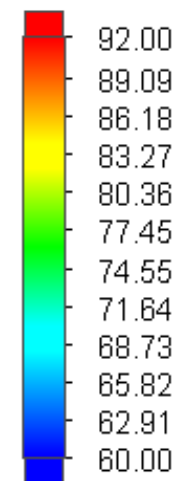
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Busbar Temperature Contours

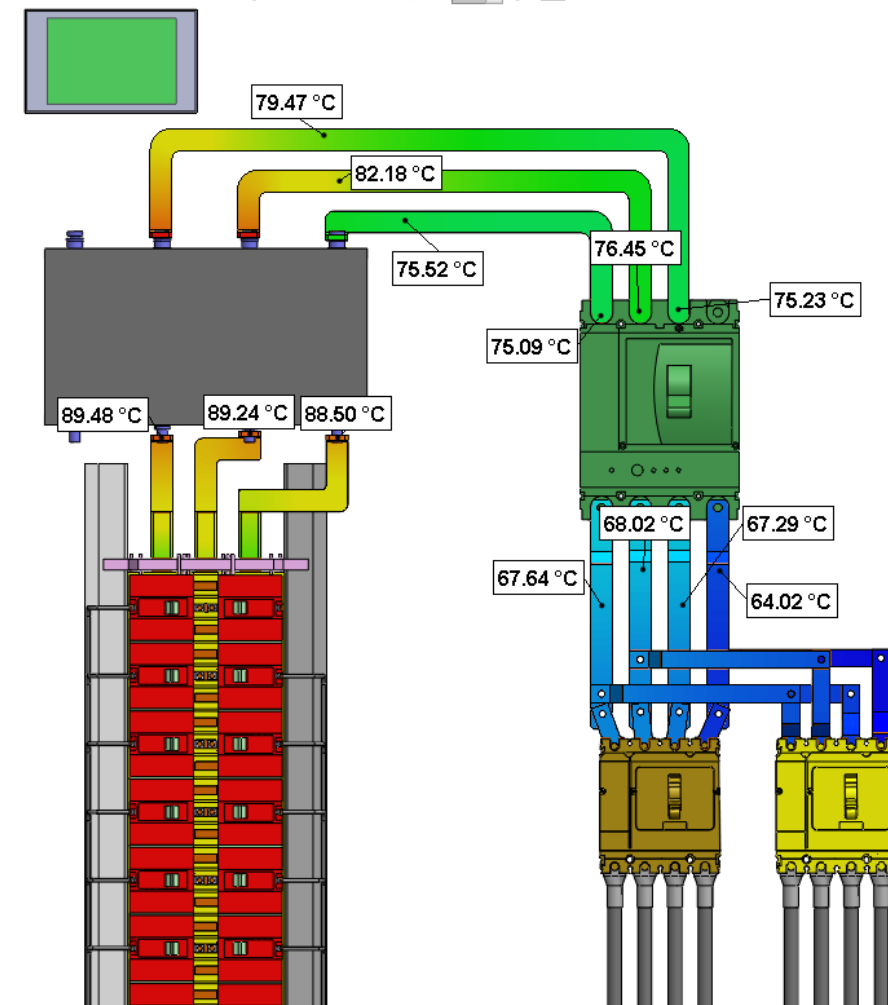
Major temperature drop can be noted in busbars of optimized design.



Base Design



Temperature (Solid) [°C]



Optimized Design

CASE STUDY | Simulation Results

Thermal Results Comparison



Parameters	Base Design	Optimized Design
Enclosure Surface Temp (Max)	84.2	71.1
Filter Max Temperature	94.4	82.6
MCCB Max Temperature	88.3	76.8
PCB Enclosure Max Temperature	96.5	90.2

Comments

The addition of side-mounted louvers, optimizing their size & location showed a significant improvement in airflow circulation and heat dissipation of both internal components & enclosure's surface temperature.



It can be concluded from the above case study that applying the suitable effective methods to the given application with their design constraints, the required design goals and a major thermal performance can be achieved.

Thank You!

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